

ECA15 TRANSMISSION LINE & WAVEGUIDES

EXP 3

Designing a Quarter wave transformer by varying following parameters

a) frequency

b) input impedance

c) characteristic impedance

Test case :

clc;

clear;

// Given values

Zr = 100;

Zs = 50;

s = 9;

d = 0.1;

r = d/2;

// Calculations

Ro = sqrt(Zs * Zr);

f = 500 * 10⁶; *// Frequency = 500 MHz*

Zof = 276 * log10(s/r);

Zoq = sqrt(Zr * Zof);

L = 300 / (f * 10⁻⁶); *// Length in meters*

// -----

// Table 1

// -----

disp(" ");

disp("=====");

```

disp("      IMPEDANCE CALCULATION TABLE");
disp("=====");
mprintf("| %-10s | %-10s | %-10s |\n","ZS (ohm)","ZR (ohm)","Ro (ohm)");
disp("-----");
mprintf("| %-10.2f | %-10.2f | %-10.2f |\n", Zs, Zr, Ro);
disp("=====");

// -----
// Table 2
// -----

disp(" ");
disp("=====");
disp("      QUARTER WAVE TRANSFORMER LENGTH");
disp("=====");
mprintf("| %-18s | %-20s |\n","Frequency (MHz)","Length (meters)");
disp("-----");
mprintf("| %-18.0f | %-20.4f |\n", f/10^6, L);
disp("=====");

// Additional Output

disp(" ");
mprintf("Characteristic impedance of quarter-wave line = %.4f ohms\n", Ro);

```

Output:

```
Scilab 2025.1.0 Console
File Edit Control Applications ?
[Icons]
Scilab 2025.1.0 Console
" "
"=====
"          IMPEDANCE CALCULATION TABLE
"=====
| ZS (ohm) | ZR (ohm) | Ro (ohm) |
"-----
| 50.00    | 100.00   | 70.71     |
"=====
" "
"=====
"          QUARTER WAVE TRANSFORMER LENGTH
"=====
| Frequency (MHz) | Length (meters) |
"-----
| 500             | 0.6000          |
"=====
" "
Characteristic impedance of quarter-wave line = 70.7107 ohms
```

Test case 2

clear;

clc;

// Given values

Zr = 100; *// Load impedance (ohms)*

f = 2000; *// Frequency in MHz (2 GHz)*

// Input impedance values

Zin = [25 50 75 100];

// Length of quarter-wave transformer

L = 300 / f; *// meters*

disp(" ");

```

disp("Quarter-Wave Transformer Design");
disp("Fixed Frequency = 2 GHz");
disp("Load Impedance (ZR) = 100 ohms");
disp(" ");

mprintf("Length of lambda/4 transformer = %.4f meters\n\n", L);

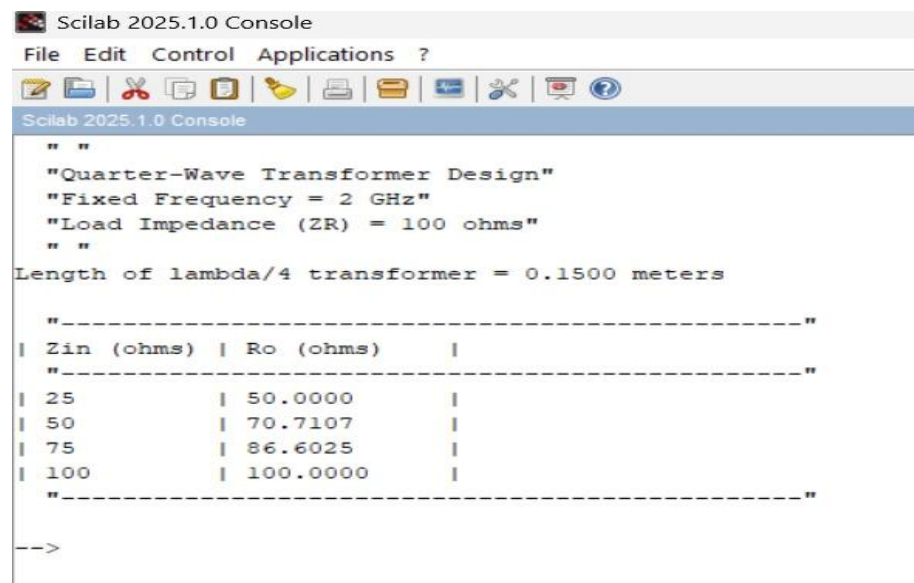
// Output Table
disp("-----");
mprintf("| %-10s | %-12s |\n", "Zin (ohms)", "Ro (ohms)");
disp("-----");

for i = 1:length(Zin)
    Ro = sqrt(Zin(i) * Zr);
    mprintf("| %-10.0f | %-12.4f |\n", Zin(i), Ro);
end

disp("-----");

```

Output



```

Scilab 2025.1.0 Console
File Edit Control Applications ?
[Icons]
Scilab 2025.1.0 Console
" "
"Quarter-Wave Transformer Design"
"Fixed Frequency = 2 GHz"
"Load Impedance (ZR) = 100 ohms"
" "
Length of lambda/4 transformer = 0.1500 meters

"-----"
| Zin (ohms) | Ro (ohms) |
"-----"
| 25          | 50.0000   |
| 50          | 70.7107   |
| 75          | 86.6025   |
| 100         | 100.0000  |
"-----"

-->

```